CVS-RFFI: Protocol-Governed Cross-Receiver Generalization and Few-Shot Incremental Registration for Spaceborne Radio-Frequency Fingerprinting

Anonymous Authors

Abstract—Radio-frequency fingerprint identification (RFFI) is evaluated with a fixed receiver, transmitter labels, and a closed class set. A spaceborne monitor violates these assumptions: the deployed receiver is absent from ground training, received IQ follows a different propagation process, and unseen transmitters must be enrolled from few labeled records under onboard memory and update constraints. We present CVS-RFFI, a protocol-governed framework that treats ground generalization and deployment registration as equal stages. Phase 1 learns asymmetric 160-dimensional identity and receiver-nuisance representations from proxy receivers under a limited-label split. Receiver-day-gated pseudo-labeling, tail-aware angular geometry, and identity-preserving challenges train the encoder before it and INT8 aggregates are sealed. Phase 2 admits only this immutable bundle and K labeled target-support records per class. Its RTB-IDR module compiles a 288-dimensional multimodal support representation into one equal-prior affine classifier using perturbation-aware robust centers, task-balanced shrinkage covariance, support-cross-fitted geometry fusion, and residual quantization. Each physical record passes once through a fixed simulated LEO channel; target queries never update the state. On a WiSig/ManySig terrestrial proxy, the selected Phase 1 model reaches 89.18% overall accuracy, 84.89% strict unseen-domain accuracy, and a 75.55% receiver floor. In a 125-row matched Phase 2 diagnostic, task-balanced covariance improves post-registration old-class accuracy by 2.62 percentage points and the old/new harmonic mean by 0.96 points for $K = 10$ with 20 new classes, but reduces new-class accuracy by 0.65 points and provides no gain at $K = 1$. The study is a simulation-based prototype, not in-orbit or flight-software validation.

Index Terms—radio-frequency fingerprint identification, specific emitter identification, cross-receiver generalization, domain adaptation, few-shot class-incremental learning, satellite communications, physical-layer authentication.

I. Introduction

Radio-frequency fingerprint identification (RFFI), also known as specific emitter identification, infers transmitter identity from hardware-dependent distortions introduced by oscillators, mixers, power amplifiers, and other nonideal radio-frequency components. Early radiometric systems used engineered transients and statistics [1]; later work showed that convolutional models can learn discriminative fingerprints directly from complex baseband samples [2]–[4]. The appeal is clear: a hardware fingerprint is harder to copy than a network identifier and can provide an independent physical-layer authentication signal.

The observable waveform is not a transmitter-only object. Propagation, receiver filtering, local oscillators, automatic gain control, sampling clocks, and preprocessing jointly transform the same emitted signal. A model can therefore achieve high same-receiver accuracy by encoding a receiver–channel shortcut rather than a stable transmitter characteristic [5], [6]. This failure is exposed when the receiver changes: cross-receiver RFFI is consistently more difficult than same-receiver classification [7]–[9].

A spaceborne monitoring lifecycle compounds receiver shift with label, channel, class, and resource constraints. Ground training cannot observe the final receiver chain, and transmitter labels are costly even in the source domains. At deployment, satellite geometry, residual synchronization errors, multipath, and noise alter the baseband process; elevation, path length, and Doppler are central non-terrestrial-link variables [10]. Short contact windows limit labeled observations, while onboard computation and storage favor ground training followed by compact inference state [11]. The class set is also open to enrollment: a newly encountered transmitter must be added from a small support set without erasing previously registered identities.

Three research lines address these difficulties separately. Cross-receiver RFFI uses collaborative receivers, adversarial disentanglement, distribution alignment, or pseudo-labeling [8], [9], [12]–[14], but normally retains a fixed transmitter set and may use target records for alignment or fine-tuning. Few-shot class-incremental SEI adds devices with prototypes, subnet expansion, self-training, or regularization [15]–[19], but its base and incremental sessions are not designed around a simultaneously unseen receiver and changed satellite-to-ground propagation process. Satellite fingerprinting establishes feasibility on real downlinks [20], [21], and ground–satellite generative learning addresses sample scarcity [11], yet these studies do not jointly evaluate replay-free old-class adaptation and new-class registration on a disjoint receiver.

The unresolved gap is therefore the intersection of the constraints, not any one constraint in isolation. Among

This manuscript is an evidence-locked initial draft. Author identities, affiliations, funding, and the final reproducibility URL are withheld pending internal approval.

the studies surveyed in Section II, we found no common evaluation that simultaneously enforces: an unseen target receiver; transmitter-label scarcity during ground learning; K-shot old- and new-class support after deployment; no runtime source IQ, clean counterpart, or target-query feedback; a compact immutable deployment state; and one role-blind predictor over all registered classes. Relaxing any one of these conditions changes the task. Target-batch alignment no longer measures source-only generalization, source replay no longer measures self-contained deployment, and separate old/new heads no longer measure a deployable all-class decision.

We ask whether an RFFI system can first learn an unseen-receiver-tolerant identity space from at most 10% transmitter-labeled source records and then adapt old identities and register new ones from K fixed received records per class, without replaying source samples or retraining on queries. This question requires two coequal stages. Phase 1 must build a representation whose transmitter geometry survives receiver and channel variation under label-limited supervision. Phase 2 must turn scarce support on the deployed receiver into a balanced old/new decision rule within a bounded state.

CVS-RFFI implements this lifecycle. Phase 1 uses an asymmetric dual-representation CV-SincNet: one branch integrates temporal, spectral, and power-amplifier evidence for transmitter identity, while a second branch absorbs receiver, channel, and noise nuisance. Receiver-day-gated semi-supervision, tail-risk geometry, and identity-preserving source challenges train the two representations before the encoder and quantized multi-sample aggregates are sealed. Phase 2 uses RTB-IDR (Robust Task-Balanced Incremental Discriminant Registration) to estimate a receiver-sensitive perturbation spectrum, robustify support centers, balance old and new tasks during covariance estimation, and compile all registered classes into one compact affine head. The channel simulator is a reproducible residual baseband operator, not evidence of an operational satellite link.

The contributions are fourfold.

1) We formalize the combined research gap as a two-stage access contract spanning label-limited source-only receiver generalization, a fixed simulated LEO satellite-to-ground observation, support-only old-class adaptation, and few-shot new-class registration under independent all-class inference.
2) For Phase 1, we develop an asymmetric identity–nuisance CV-SincNet that couples physical signal views with explicit receiver-information allocation, receiver-day-gated pseudo-labeling, tail-aware angular risk, and an identity-preserving source-domain curriculum. The output is a sealed encoder plus auditable INT8 aggregate knowledge rather than replayable source data.
3) For Phase 2, we develop RTB-IDR, a support-only compiler that combines learned and deterministic views, perturbation-aware robust centers, task-balanced shrinkage geometry, bounded Fisher residuals, and residual quantization in one equal-prior affine classifier.
4) We report matched positive and negative evidence for both stages. A 32-candidate Phase 1 audit identifies a source-only checkpoint, while a complete 125-row Phase 2 diagnostic shows that task balancing reduces old-class forgetting at large registration scales but does not solve one-shot adaptation or cross-role calibration.

The remainder of the paper reviews related work, formalizes the protocol, describes the two-stage method, reports the evidence currently available, and lists the experiments required before submission-grade performance claims can be made.

II. Related Work

A. Deep RFFI and Channel Robustness

Deep RFFI has progressed from raw-IQ convolutional classifiers [2]–[4] to models that use multisampling [22], spectrograms [23], or learned channel compensation [5]. ORACLE explicitly optimizes transmitter classification with convolutional networks [4], whereas the channel-dissection study of Al-Shawabka et al. shows that an apparently accurate classifier can learn propagation artifacts rather than stable hardware characteristics [6]. WiSig provides a large-scale test bed with 174 transmitters, 41 receivers, and multiple capture campaigns [7]. We use it only as a terrestrial proxy; the present experiments are not measurements from an on-orbit receiver.

B. Cross-Receiver Generalization and Adaptation

Receiver mismatch has prompted several complementary approaches. Collaborative receiver-agnostic RFFI exploits multiple receivers [8]. RIEI explicitly separates emitter- and receiver-related information and also studies a federated variant [9]. Other methods use target-domain alignment and pseudo-labeling [12], dynamic distribution alignment [13], or joint channel-robust and receiver-independent learning [14]. These studies build on broader DG and domain-adversarial learning principles [24]–[26].

Our distinction is primarily one of lifecycle and evidence. Phase 1 permits only source receivers. Phase 2 permits labeled target support but excludes target-query feedback and runtime source samples. Consequently, a method that uses target data during model selection is classified as adaptation, not DG; a comparison method that requires source replay remains a valid external comparator but is reported under a different permission row.

C. Satellite Transmitter Fingerprinting

PAST-AI and SatIQ provide evidence that satellite transmitters can be fingerprinted from real received traffic [20], [21]. A recent ground–satellite generative framework further studies few-shot satellite IoE identification [11]. Their real-satellite evidence is important context, but it cannot be transferred to a proxy experiment. In CVS-RFFI, WiSig/ManySig supplies terrestrial device and receiver observations. A simulated LEO residual channel then creates clear-sky, low-elevation, and rainy-link baseband profiles. We therefore use "spaceborne deployment proxy" rather than "in-orbit validation" throughout.

D. Few-Shot and Class-Incremental SEI

Few-shot metric learning commonly represents each class by a support prototype [27]. Class-incremental learning also addresses catastrophic forgetting, often through exemplars, distillation, or classifier expansion [28], [29]. In wireless identification, CSIL expands a device classifier while preserving old knowledge [15]; later methods add self-training and prototype augmentation [16], systematic few-shot incremental design [17], orthogonal or reserved classifier directions [18], and prototype correction with hierarchical regularization [19]. Open-world variants further combine incremental enrollment with rejection or semi-supervised learning [30]–[32].

These methods do not all solve the same problem. Some assume complete base-session training, retain an old teacher, or update a neural network over many gradient steps. CVS-RFFI instead studies a sealed pretrained backbone followed by support-only registration on a new receiver. Its current RTB-IDR realization uses a parametric discriminant state rather than exemplar replay or query-graph inference.

III. Problem Formulation and Protocol

A. Compound Identification Task

Let $y \in \mathcal{Y}$ denote a transmitter identity, $d \in \mathcal{D}$ a receiver domain, and $c \in \mathcal{C}$ a propagation condition. We write a length-N complex baseband observation as

$$\mathbf{x}_{y,d,c} = \mathcal{R}_d[\mathcal{H}_c(\mathcal{T}_y(\mathbf{s}))] + \mathbf{n}, \tag{1}$$

where $\mathcal{T}_y$ represents transmitter-dependent hardware distortion, $\mathcal{H}_c$ propagation and synchronization effects, $\mathcal{R}_d$ the receiver chain, and $\mathbf{n}$ additive noise. The objective is to preserve information about y while suppressing shortcuts tied to d and incidental c. Equation (1) is a task decomposition rather than an identifiable inverse model: CVS-RFFI neither reconstructs clean IQ nor estimates a physical power-amplifier model at deployment. Memory-polynomial analysis [33] motivates stable nonlinear transmitter effects, but the classifier operates on the received record.

The deployment problem combines domain shift and class-set expansion. Source training observes receivers $\mathcal{R}_{\mathrm{s}}$ and old identities $\mathcal{Y}_{\mathrm{o}}$. Deployment observes a disjoint receiver set $\mathcal{R}_{\mathrm{t}}$, retains the old classes, and registers $\mathcal{Y}_{\mathrm{n}}$, where

$$\mathcal{R}_{\mathrm{s}} \cap \mathcal{R}_{\mathrm{t}} = \varnothing, \qquad \mathcal{Y}_{\mathrm{o}} \cap \mathcal{Y}_{\mathrm{n}} = \varnothing. \tag{2}$$

The final predictor is a single map over $\mathcal{Y}_{\mathrm{o}} \cup \mathcal{Y}_{\mathrm{n}}$; it is not given an old/new query role. This distinguishes the task from closed-set DG alone, target adaptation alone, and conventional class-incremental learning under source replay.

B. Simulated LEO Satellite-to-Ground Channel

The simulator is applied to a captured WiSig/ManySig record, not to a clean transmitted waveform. It represents a post-synchronization residual satellite-to-ground baseband channel. The three manuscript-facing profiles are

$$\mathcal{C}_{\mathrm{LEO}} = \{\mathrm{CS}, \mathrm{LE}, \mathrm{RL}\}, \tag{3}$$

denoting clear-sky, low-elevation, and rainy-link conditions, respectively. Given $\mathbf{x}_i = [x_i[0], \ldots, x_i[N-1]]$, the implemented operator is

$$u_i[n] = \sum_{\ell=0}^{1} h_{i,\ell} x_i[n - d_{i,\ell}], \tag{4}$$

$$q_i[n] = u_i[n] \exp\left\{ j \left(\frac{2\pi \Delta f_i n}{f_{\mathrm{s}}} + \phi_i[n] \right) \right\}, \tag{5}$$

$$\widetilde{x}_i[n] = \frac{a_i q_i[n]}{\sqrt{\max\left\{ N^{-1} \sum_{m=0}^{N-1} |q_i[m]|^2, \epsilon \right\}}} + w_i[n]. \tag{6}$$

Here, $d_{i,0} = 0$, samples with negative indices are zero, the second delay is sampled from a bounded integer interval, $a_i = 10^{g_i/20}$ is a residual AGC gain, and $\epsilon = 10^{-12}$. The noise obeys $w_i[n] \sim \mathcal{CN}(0, \sigma_{w,i}^2)$, with $\sigma_{w,i}^2 = P_i 10^{-\gamma_i/10}$, where P_i is the post-AGC signal power and γ_i the sampled SNR in decibels. The phase process is Wiener:

$$\phi_i[n] = \phi_i[n-1] + \varepsilon_i[n], \qquad \varepsilon_i[n] \sim \mathcal{N}(0, \sigma_{\phi,i}^2), \tag{7}$$

and the residual carrier offset follows

$$\Delta f_i \sim \mathcal{N}(0, \sigma_{f,c_i}^2). \tag{8}$$

Bulk orbital Doppler is assumed to have been removed by synchronization; the simulator injects only Δf_i.

The dominant tap is Rician or shadowed Rician [34]:

$$h_{i,0} = \sqrt{\rho_0} \left[\eta_i \sqrt{\frac{K_i}{K_i + 1}} e^{j\psi_i} + \sqrt{\frac{1}{K_i + 1}} \, z_{i,0} \right], \tag{9}$$

$$h_{i,1} = \sqrt{\rho_1} \, z_{i,1}, \tag{10}$$

where $z_{i,0}, z_{i,1} \sim \mathcal{CN}(0,1)$, $\psi_i \sim \mathcal{U}(-\pi, \pi)$, and $(\rho_0, \rho_1) = (1, p_c)/(1 + p_c)$. For the low-elevation profile, $\eta_i = 10^{v_i/20}$ with $v_i \sim \mathcal{N}(0, 1.5^2)$ dB; otherwise $\eta_i = 1$. The Rician factor varies linearly with elevation:

$$K_i^{\mathrm{dB}} = K_{\min,c_i} + (K_{\max,c_i} - K_{\min,c_i}) \frac{\theta_i - \theta_{\min,c_i}}{\theta_{\max,c_i} - \theta_{\min,c_i}}. \tag{11}$$

The gain equations use $K_i = 10^{K_i^{\mathrm{dB}}/10}$.

Following standard LEO geometry [10], altitude h_i and elevation θ_i determine the slant range

$$d_i = -R_{\mathrm{E}} \sin\theta_i + \sqrt{(R_{\mathrm{E}} + h_i)^2 - R_{\mathrm{E}}^2 \cos^2\theta_i} \tag{12}$$

and free-space path loss

$$L_{\mathrm{FS},i} = 20 \log_{10} \left(\frac{4\pi d_i f_{\mathrm{c}}}{c_0} \right). \tag{13}$$

Here, $R_{\mathrm{E}} = 6.371 \times 10^6$ m and $c_0 = 299\,792\,458$ m/s. The present residual-channel implementation records d_i and $L_{\mathrm{FS},i}$ as metadata but does not multiply absolute path loss

TABLE I
Common Simulated LEO Channel Settings

Parameter	Setting
Record length N	256 complex samples
Sampling rate f_{s}	25 MS/s
Carrier frequency f_{c}	2.462 GHz
Orbit and altitude h	LEO; $\mathcal{U}(500, 2000)$ km
Multipath	Two taps; first delay zero
Bulk orbital Doppler	Compensated before residual model
Absolute FSPL in IQ	Not applied; retained as metadata
Explicit atmospheric attenuation	Disabled
Additional IQ imbalance	Disabled
AGC target RMS	1, followed by residual gain g
Realization policy	One seeded, fixed output per physical record

into IQ, because every record is RMS-normalized by the modeled AGC. Likewise, explicit atmospheric attenuation and additional IQ imbalance are disabled. The rainy-link profile therefore represents the post-AGC SNR, fading, phase-noise, and residual-frequency regime in Table II; it is not a full link-budget rain-attenuation model.

C. Phase 1: Source-Only Learning

Phase 1 uses the labeled source set

$$\mathcal{L}_{\mathrm{s}} = \{(\mathbf{x}_i, y_i, d_i) : d_i \in \mathcal{R}_{\mathrm{s}}\} \tag{14}$$

and the unlabeled source set

$$\mathcal{U}_{\mathrm{s}} = \{(\mathbf{x}_j, d_j) : d_j \in \mathcal{R}_{\mathrm{s}},\ y_j \text{ hidden}\}. \tag{15}$$

The protocol limits the transmitter-labeled fraction to $\rho_{\mathrm{label}} \leq 0.10$; its current governed split allocates 0.07, 0.63, and 0.30 of the source pool to labeled training, unlabeled training, and disjoint source validation. No record from $\mathcal{R}_{\mathrm{t}}$ may affect training, normalization statistics, early stopping, thresholds, prototypes, or checkpoint selection. Phase 1 outputs an immutable bundle $\mathcal{B}_{\mathrm{P1}}$ containing the selected encoder and licensed aggregate state, but no target-derived quantity. The historical checkpoint audited in Section V-B used an earlier 0.10/0.70/0.20 split and is therefore internal method evidence rather than the final protocol-locked confirmation.

D. Phase 2: Fixed Observation and Support-Only State

Before Phase 2, each physical IQ record produces exactly one allowed simulated LEO observation

$$\widetilde{\mathbf{x}}_i = \mathcal{H}_{\mathrm{LEO}}(\mathbf{x}_i; c_i, \boldsymbol{\xi}_i)\,, \qquad c_i \in \mathcal{C}_{\mathrm{LEO}}, \tag{16}$$

where $\boldsymbol{\xi}_i$ is the seeded random state governing the parameters in Tables I and II. The scenario assignment, seed, and resulting bytes are frozen before support/query use. Equalization, normalization, FFTs, and other mathematical views may read $\widetilde{\mathbf{x}}_i$, but they do not create additional physical samples. Thus, K-shot means K distinct physical records, not K transformations of one record.

Let the old and newly enrolled class sets be $\mathcal{Y}_{\mathrm{o}}$ and $\mathcal{Y}_{\mathrm{n}}$, with $\mathcal{Y}_{\mathrm{o}} \cap \mathcal{Y}_{\mathrm{n}} = \varnothing$. Every registered class $c \in \mathcal{Y} = \mathcal{Y}_{\mathrm{o}} \cup \mathcal{Y}_{\mathrm{n}}$ has a labeled support set

$$\mathcal{S}_c = \{(\widetilde{\mathbf{x}}_{c,k}, c)\}_{k=1}^{K}. \tag{17}$$

Support and query physical identifiers are disjoint. A deployment state is compiled as

$$\Theta = \mathrm{Compile}\left(\mathcal{B}_{\mathrm{P1}}, \bigcup_{c\in\mathcal{Y}} \mathcal{S}_c, \Gamma\right), \tag{18}$$

where $\mathcal{B}_{\mathrm{P1}}$ is an immutable Phase 1 bundle and Γ is a method configuration frozen before query evaluation.

For every query, the predictor produces one independent decision

$$\widehat{y}_j = \arg\max_{c\in\mathcal{Y}} s_c(\widetilde{\mathbf{x}}_j^{(q)}; \Theta). \tag{19}$$

The predictor cannot read the query label, its true old/new role, the number of examples from each class in the query batch, or another query's features. It cannot enforce class quotas or use Hungarian assignment, optimal transport, label propagation over queries, or global reassignment. Labels are joined only after immutable predictions are emitted.

E. Metrics

Let $A_{\mathrm{o}}^{\mathrm{pre}}$ be old-class accuracy after Stage 2-B and before adding new classes, $A_{\mathrm{o}}^{\mathrm{post}}$ old-class accuracy after Stage 2-C, and A_{n} seen-new accuracy. We report

$$F = A_{\mathrm{o}}^{\mathrm{pre}} - A_{\mathrm{o}}^{\mathrm{post}} \tag{20}$$

and

$$H_{\mathrm{old,new}} = \frac{2A_{\mathrm{o}}^{\mathrm{post}} A_{\mathrm{n}}}{A_{\mathrm{o}}^{\mathrm{post}} + A_{\mathrm{n}}}. \tag{21}$$

We also report the lowest old-class accuracy within each row, receiver-wise results, and scenario-wise results. A valid Stage 2-C claim must keep $A_{\mathrm{o}}^{\mathrm{post}}$, A_{n}, and $H_{\mathrm{old,new}}$ from the same experimental row.

IV. Proposed CVS-RFFI Framework

A. Lifecycle Overview

CVS-RFFI does not treat Phase 1 as generic pre-training or Phase 2 as an auxiliary classifier. Phase 1 determines which transmitter evidence remains stable when the receiver changes; Phase 2 determines how that evidence can be recalibrated and extended when only target support is available. The target query is absent from both state-construction processes. Figure 1 shows the resulting ground-to-deployment boundary.

B. Phase 1: Label-Limited Cross-Receiver Representation Learning

1) Asymmetric Physical Multi-View Encoder: The frozen Phase 1 model receives $\mathbf{x} \in \mathbb{R}^{2\times256}$. A shared front end applies 24 learnable Sinc bandpass filters of length

TABLE II
Scenario-Specific Parameters of the Simulated Residual LEO Channel

Parameter	Clear-sky (CS)	Low-elevation (LE)	Rainy-link (RL)	Sampling rule / unit
Elevation θ	35–90	10–35	20–80	Uniform, degrees
SNR	22–32	16–28	14–26	Uniform, dB
Residual CFO σ_f	50	90	70	Gaussian std., Hz
Phase-increment std. σ_ϕ	0–$5{\times}10^{-4}$	$1{\times}10^{-4}$–$8{\times}10^{-4}$	$1{\times}10^{-4}$–$7{\times}10^{-4}$	Uniform, rad/sample
Fading law	Rician	Shadowed Rician	Rician	LOS shadow std. in LE: 1.5 dB
K^{dB} range	16–24	8–18	10–20	Linear in elevation
Second-tap delay d_1	1–2	1–3	1–3	Discrete uniform, samples
Second-tap factor p_c	0.08	0.12	0.10	Powers normalized as $[1, p_c]$
Residual AGC g	−0.2–0.2	−0.3–0.3	−0.3–0.3	Uniform, dB

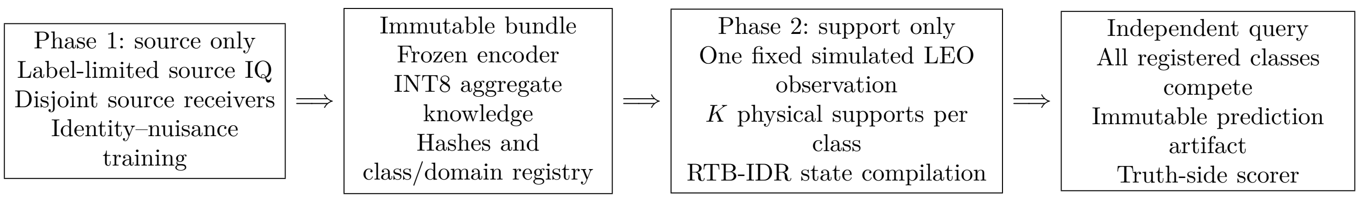

Fig. 1. The CVS-RFFI lifecycle. The target query is absent from training and registration. The independent scorer receives ground truth only after predictions are sealed.

TABLE III
Phase 2 Access and Claim Semantics

Stage	Target information	Permitted claim
2-A	No target transmitter labels; optionally unlabeled fixed received IQ	Zero-label target reference only
2-B	K-shot labeled support for $\mathcal{Y}_{\mathrm{o}}$	Old-class target adaptation and calibration
2-C	K-shot labeled support for $\mathcal{Y}_{\mathrm{o}} \cup \mathcal{Y}_{\mathrm{n}}$	Joint old-class retention and seen-new enrollment

79 [35] and first- and second-order high-frequency differences. Above this stem, two parameter-distinct backbones produce

$$\mathbf{z}_{\mathrm{id}} = f_\theta^{\mathrm{id}}(\mathbf{x}) \in \mathbb{R}^{160}, \qquad \mathbf{z}_{\mathrm{dom}} = f_\theta^{\mathrm{dom}}(\mathbf{x}) \in \mathbb{R}^{160}. \tag{22}$$

The identity backbone combines temporal, spectral, and power-amplifier (PA) evidence. Its PA path explicitly lifts the received IQ through a short memory polynomial [33],

$$\begin{aligned}\phi_{r,o}(\mathbf{x})[n] &= x[n-r]|x[n-r]|^{o-1},\\ &r \in \{0,1,2,3\}, \qquad o \in \{1,3,5\},\end{aligned} \tag{23}$$

before gated convolution and projection. Temporal, spectral, circularity, and PA features are fused into the final identity vector, which is the only learned embedding used for Phase 2 class geometry.

The domain backbone is deliberately asymmetric. It disables style mixing, retains a widely linear complex path for DAC/IQ conjugate coupling, and augments its learned domain feature with 18 receiver–channel–noise statistics. Let $\mathbf{r}_{\mathrm{rcn}}(\mathbf{x})$ collect these statistics and let $\mathbf{z}^0_{\mathrm{dom}}$ denote the learned domain feature. The implemented gated enhancement is

$$\mathbf{e}_{\mathrm{rcn}} = E_{\mathrm{rcn}}(\mathbf{r}_{\mathrm{rcn}}(\mathbf{x})), \tag{24}$$

$$\mathbf{g}_{\mathrm{rcn}} = \sigma\big(\mathbf{W}_g[\mathbf{z}^0_{\mathrm{dom}}; \mathbf{e}_{\mathrm{rcn}}] + \mathbf{b}_g\big), \tag{25}$$

$$\mathbf{z}_{\mathrm{dom}} = \mathrm{LN}\big(\mathbf{z}^0_{\mathrm{dom}} + 0.35\, \mathbf{g}_{\mathrm{rcn}} \odot \mathbf{e}_{\mathrm{rcn}}\big). \tag{26}$$

Conversely, the identity branch removes the DAC path and applies same-transmitter cross-domain MixStyle [25] only to labeled temporal features. Training executes both branches; deployment bypasses the domain branch. This construction allocates nuisance evidence to a separate path, but the shared low-level stem means that statistical independence is neither assumed nor claimed.

2) Receiver-Nuisance Allocation: The identity head uses CosFace [36]. For transmitter class c, its frozen logit is

$$\ell^{\mathrm{tx}}_{i,c} = 30 \left[\frac{(\mathbf{z}^{\mathrm{id}}_i)^{\top} \mathbf{w}_c}{\|\mathbf{z}^{\mathrm{id}}_i\|_2 \|\mathbf{w}_c\|_2} - 0.35\, \mathbf{1}[c = y_i] \right], \tag{27}$$

and the supervised transmitter loss uses 0.01 label smoothing. Receiver/day domain labels then play two opposite roles. A domain head predicts d_i from $\mathbf{z}_{\mathrm{dom}}$, whereas a gradient-reversal head [24] predicts d_i from $\mathbf{z}_{\mathrm{id}}$ while reversing the gradient entering the identity backbone:

$$\mathcal{L}_{\mathrm{dom}} = -\frac{1}{B} \sum_i \log p_\phi(d_i \mid \mathbf{z}^{\mathrm{dom}}_i), \tag{28}$$

$$\mathcal{L}_{\mathrm{adv}} = -\frac{1}{B} \sum_i \log p_\psi\big(d_i \mid \mathrm{GRL}(\mathbf{z}^{\mathrm{id}}_i)\big). \tag{29}$$

Two additional terms prevent the two heads from satisfying their labels through unrelated shortcuts. For centered batch matrices $\widetilde{\mathbf{Z}}_{\text{id}}$ and $\widetilde{\mathbf{Z}}_{\text{dom}}$,

$$\begin{aligned}\mathcal{L}_{\text{orth}} &= \frac{1}{160^2}\left\|\frac{\widetilde{\mathbf{Z}}_{\text{id}}^{\mathsf{T}}\widetilde{\mathbf{Z}}_{\text{dom}}}{B-1}\right\|_{\text{F}}^2, \\ \mathcal{L}_{\text{cons}} &= \frac{1}{|\mathcal{P}|}\sum_{(c,d,d')\in\mathcal{P}}\left[1-\cos(\boldsymbol{\mu}_{c,d},\boldsymbol{\mu}_{c,d'})\right],\end{aligned} \tag{30}$$

where $\boldsymbol{\mu}_{c,d}$ is the normalized identity center of transmitter c in source domain d, and $\mathcal{P}$ contains available same-transmitter, different-domain center pairs. The scheduled separation loss is

$$\begin{aligned}\mathcal{L}_{\text{sep}}(e) &= w_{\text{dom}}(e)\mathcal{L}_{\text{dom}} + w_{\text{adv}}(e)\mathcal{L}_{\text{adv}} \\ &\quad + w_{\text{orth}}(e)\mathcal{L}_{\text{orth}} + w_{\text{cons}}(e)\mathcal{L}_{\text{cons}}.\end{aligned} \tag{31}$$

Its maximum weights are 1, 0.35, 0.05, and 0.08, respectively. Domain observability, adversarial suppression, cross-covariance reduction, and class-conditional alignment are thus optimized together; none alone proves complete disentanglement.

3) Receiver-Day-Gated Semi-Supervision and Tail Risk: In the audited checkpoint, 10% of the source pool carries transmitter labels, while 70% is available without transmitter labels and 20% is reserved for source validation. During epochs 131–200, an exponential-moving-average (EMA) teacher with decay 0.999 predicts a lightly perturbed unlabeled record. For confidence $\kappa_j = \max_c q_{j,c}$, the receiver-day threshold is

$$\tau_d = \operatorname{clip}(Q_{0.86}\{\kappa_j : d_j = d\},\, 0.92,\, 0.97)\,. \tag{32}$$

A pseudo-label is admitted only if it clears this threshold, the teacher's domain prediction agrees with the known source domain, the adjacent time-window decision is consistent, and the student agrees under a stronger perturbation. With the three agreement indicators denoted by $g_j^{\text{dom}}, g_j^{\text{temp}}, g_j^{\text{str}}$,

$$m_j = \mathbf{1}[\kappa_j \ge \tau_{d_j}]g_j^{\text{dom}}g_j^{\text{temp}}g_j^{\text{str}}, \tag{33}$$

$$\mathcal{L}_u = -\frac{\sum_j m_j \log p_\theta(\widehat{y}_j \mid a_s(\mathbf{u}_j))}{\sum_j m_j + \epsilon}. \tag{34}$$

An entropy term is evaluated on all strong-view predictions. Receiver-day conditioning prevents an easy domain from setting the acceptance pattern for every receiver; the gates control label noise rather than increasing the nominal number of labeled records.

Mean source accuracy can still hide a collapsed receiver or transmitter. For each source domain g, the model computes its label-smoothed transmitter loss ℓ_g and averages only the largest 35%:

$$\mathcal{L}_{\text{group}} = \frac{1}{K_g}\sum_{g\in\operatorname{TopK}_{K_g}(\{\ell_h\})}\ell_g, \qquad K_g = \lceil 0.35|\mathcal{G}|\rceil\,. \tag{35}$$

A domain-wise logit-gradient-variance penalty aligns optimization variability across source domains. The identity space is further constrained by cross-epoch prototypes, cross-domain supervised contrast, a 40° angular radius, class-balanced tail risk, core–tail preservation, and soft interclass boundary mixtures. These are closed-set source-domain objectives; they do not constitute unknown-class training.

TABLE IV
Phase 1 Optimization Schedule

Epochs	Data stage	Dominant addition
1–16	Labeled	Core TX/domain learning; partial adversarial, orthogonal, and hard-domain weights
17–68	Labeled	Ramp receiver suppression, class-conditional alignment, and tail geometry
69–79	Labeled	Full separation and source-domain risk weights
80–130	Labeled	Add simulated LEO transmitter classification
131–200	Labeled + unlabeled	Add gated pseudo-label CE and entropy minimization

4) Identity-Preserving Source Challenges: Phase 1 creates source-only challenges for which the transmitter label must remain invariant. Same-transmitter cross-domain MixStyle starts with probability 0.18, shape parameter 0.10, and strength 0.70; after epoch 110, probability and strength anneal over 40 epochs to 0.05 and 0.32. A leave-one-source-domain episode constructs an explicit held-domain classification problem within each labeled batch. From epoch 80, labeled records also receive a simulated residual LEO channel view and retain their original transmitter label:

$$\mathcal{L}_{\text{LEO}} = -\frac{1}{B}\sum_i \log p_\theta(y_i \mid a_{\text{LEO}}(\mathbf{x}_i))\,. \tag{36}$$

The active coefficient is 0.68. The frozen checkpoint assigns zero weight to clean-to-LEO consistency; only the classification loss in (36) contributes a gradient. MixStyle, episodic DG, and LEO augmentation are established mechanisms individually. Their role here is the identity-preserving coordination of receiver style, held-source extrapolation, and deployment-channel stress.

5) Scheduled Objective and Immutable Bundle: For readability, the implemented nonzero terms are grouped as

$$\begin{aligned}\mathcal{L}_{\text{P1}}(e) &= \mathcal{L}_{\text{tx}} + \mathcal{L}_{\text{sep}}(e) + \mathcal{L}_{\text{risk}}(e) + \lambda_{\text{epi}}(e)\mathcal{L}_{\text{epi}} \\ &\quad + 0.68\,\mathbf{1}[e\ge 80]\mathcal{L}_{\text{LEO}} \\ &\quad + \mathbf{1}[e\ge 131]\,(0.16\mathcal{L}_u + 0.01\mathcal{L}_{\text{ent}})\,.\end{aligned} \tag{37}$$

Here, $\mathcal{L}_{\text{risk}}$ groups the nonzero hard-domain, gradient-variance, prototype, angular, tail, and soft-boundary terms described above; their individual coefficients and warm-up epochs are frozen before target access. Table IV summarizes the dominant schedule. Style mixing acts inside the identity feature path and is therefore not written as an additive loss.

Checkpoint selection uses only disjoint source validation and a frozen joint criterion over overall, strict unseen-domain, and receiver-floor behavior. The resulting deployment bundle contains the checkpoint, architecture and normalization metadata, class/domain registries, and 84 quantized domain–class aggregate identity centers. Each center is formed from at least two distinct Phase 1 physical records before target access and stored with an INT8 code, scale, and validity mask. The bundle contains no raw source IQ, single-sample feature, reversible index, or full-precision exemplar. Phase 2 reads but cannot update it. Changing this aggregate state changes the bundle identifier, not the already validated target-data capsule.

C. Phase 2: Support-Only Incremental Registration

Phase 2 addresses the complementary deployment problem with the Phase 1 encoder and bundle frozen. RTB-IDR receives labeled target support for all currently registered old and new classes, performs no gradient update, and compiles a single state used by every subsequent query. The procedure cannot access source records, clean counterparts, query features from other decisions, or query truth.

1) Multimodal Target Feature: For each fixed received record, RTB-IDR computes three views from the same physical observation:

$$\mathbf{a} = \mathcal{N}_\epsilon(\mathbf{z}_{\mathrm{id}}) \in \mathbb{R}^{160}, \tag{38}$$

$$\mathbf{b} = \mathcal{N}_\epsilon\left([\mathbf{f}_{\mathrm{FFT96}}; \mathbf{f}_{\mathrm{RF32}}]\right) \in \mathbb{R}^{128}, \tag{39}$$

where $\mathcal{N}_\epsilon(\mathbf{v}) = \mathbf{v}/\max(\|\mathbf{v}\|_2, \epsilon)$ and $\epsilon = 10^{-8}$. FFT96 is obtained by centering, applying a Hann window, taking the log-magnitude spectrum, and interpolating to 96 bins. RF32 contains fixed amplitude, phase, moment, imbalance, and short-lag correlation statistics. Neither view increases K.

The joint feature is

$$\mathbf{z} = \mathcal{N}_\epsilon\left(\begin{bmatrix}\mathbf{a} \\ 4\mathbf{b}\end{bmatrix}\right) \in \mathbb{R}^{288}. \tag{40}$$

For nonzero blocks, their final norm contributions are $1/\sqrt{17}$ and $4/\sqrt{17}$. The fixed factor 4 is part of the locked candidate and is not selected from target-query accuracy.

2) Perturbation-Aware Robust Centers: The INT8 source aggregates are dequantized, centered within each class across source domains, and pooled into a perturbation covariance $\mathbf{G}$. After symmetric correction and subtraction of the estimated quantization-noise floor,

$$\mathbf{G}_+ = \frac{\mathbf{G} + \mathbf{G}^\top}{2} - \sigma_{\mathrm{q}}^2 \mathbf{I}. \tag{41}$$

The positive eigenspectrum $(\lambda_j, \mathbf{u}_j)$ defines a class-agnostic basis $\mathbf{U}$. Its rank is the ceiling of the positive-spectrum participation ratio, so rank selection does not use target-query results.

For class c, let $\bar{\mathbf{z}}_c^{\mathrm{id}}$ be the mean identity support and $\mathbf{e}_{c,k}$ the residual of support k. The perturbation energy is

$$E_{c,k} = \sum_{j=1}^{r} \rho_j \left(\mathbf{u}_j^\top \mathbf{e}_{c,k}\right)^2, \qquad \rho_j = \frac{\lambda_j}{\sum_{\ell=1}^{r} \lambda_\ell}. \tag{42}$$

With $\tau_c = K^{-1} \sum_k E_{c,k}$, one-step Cauchy weights are

$$\omega_{c,k} \propto \frac{1}{1 + E_{c,k}/\tau_c}. \tag{43}$$

The weighted identity center defines a translation applied uniformly to all identity supports of class c; their within-class residuals and the FFT/RF blocks are unchanged. For $K \le 2$ or a degenerate energy scale, the module takes a conservative identity fallback. This prevents the method from fabricating covariance evidence at one shot.

3) Task-Balanced Shrinkage Geometry: The 288-dimensional support covariance is ill-conditioned when K is small. RTB-IDR therefore uses automatic shrinkage [37] separately within the old and new tasks:

$$\widehat{\boldsymbol{\Sigma}}_{\mathrm{o}} = \mathrm{ShrinkCov}(\{\mathbf{z}_{c,k} : c \in \mathcal{Y}_{\mathrm{o}}\}), \tag{44}$$

$$\widehat{\boldsymbol{\Sigma}}_{\mathrm{n}} = \mathrm{ShrinkCov}(\{\mathbf{z}_{c,k} : c \in \mathcal{Y}_{\mathrm{n}}\}). \tag{45}$$

After registration, the balanced covariance is

$$\widehat{\boldsymbol{\Sigma}}_{\mathrm{bal}} = \frac{1}{2}\widehat{\boldsymbol{\Sigma}}_{\mathrm{o}} + \frac{1}{2}\widehat{\boldsymbol{\Sigma}}_{\mathrm{n}}. \tag{46}$$

The 0.5/0.5 weights are fixed and do not vary with the number of new classes, receiver, scenario, seed, or transmitter identifier. Before new-class registration, and in the $K \le 2$ fallback, the registered-old control geometry is retained exactly.

Two covariance structures are constructed. A full model retains cross-block correlations among identity, FFT, and RF features; a block model zeros the cross-block terms. Leave-one-shot-rank cross-fitting estimates class-wise reliability without query access. The resulting fusion is compiled into a single score matrix; it is not a query-time mixture chosen from a query's old/new role.

4) Bounded Fisher Residual and Affine Compilation: A low-rank Fisher residual [38] is estimated from support class means and within-class scatter. Candidate residuals are clipped and admitted only through support-cross-fitted per-class and atomic safety checks. Failed candidates leave the base geometry unchanged. Equal class priors are then used to compile a linear discriminant head

$$\mathbf{w}_c = \widehat{\boldsymbol{\Sigma}}^{-1} \boldsymbol{\mu}_c, \tag{47}$$

$$b_c = -\frac{1}{2} \boldsymbol{\mu}_c^\top \widehat{\boldsymbol{\Sigma}}^{-1} \boldsymbol{\mu}_c, \tag{48}$$

$$s_c(\mathbf{q}) = \mathbf{q}^\top \mathbf{w}_c + b_c. \tag{49}$$

All old and new classes use the same score form and equal prior. Thus, task balancing affects covariance estimation but does not create separate old/new heads.

5) Quantized State: Each affine coefficient block is stored with two residual INT8 codes and two FP16 scales; intercepts use FP16 and the diagonal metric uses FP32. For C registered classes and $p = 288$,

$$B_{\mathrm{core}} = 4p + 2Cp + 14C \text{ bytes}. \tag{50}$$

The 26-class core arrays occupy 16.11 KiB, and the compiled head requires $Cp = 7488$ multiply–accumulate operations per query. These figures exclude the frozen encoder, FFT/RF extraction, registration workspace, metadata, and the current FP32 decode path.

V. Experimental Methodology

A. Data and Splits

We use the WiSig/ManySig corpus as a terrestrial receiver/transmitter proxy. The public WiSig dataset contains 10 million packets from 174 transmitters and 41 receivers across four captures [7]. The frozen CVS-RFFI subset uses seven source receivers

$$\{1-1, 1-19, 14-7, 18-2, 19-2, 2-1, 2-19\}$$

and five disjoint target receivers

$$\{20-1, 3-19, 7-14, 7-7, 8-8\}.$$

Six transmitter identities appear in Phase 1 and are old classes in Phase 2. Nested sets of 5, 10, and 20 additional identities are used as new classes. Every target record is assigned one of $\{\mathrm{CS}, \mathrm{LE}, \mathrm{RL}\}$ before support/query selection. The received record length, sampling rate, carrier frequency, geometry, fading, synchronization residuals, and noise ranges are fixed by Tables I and II.
[AUTHOR ACTION: Insert the exact public data-release identifier, local manifest hash, and per-split physical-record counts from the frozen capsule.]

B. Phase 1 Selection Audit

The Phase 1 audit contains 32 completed candidates. Each has a complete 200–240 epoch trace, checkpoint, diagnostic export, and source-only evaluation. The selected checkpoint, ADV3B02_CORE90_SOFT_E200, uses 200 epochs and is selected at epoch 194. It is compared with the preceding ADV2 aggregate rather than with an untraceable cross-run maximum. This historical audit used the 0.10/0.70/0.20 source split implemented by that run. A final confirmation must use the current 0.07/0.63/0.30 protocol split; Table V is not relabeled as evidence from that unexecuted split.

We report overall source-domain accuracy, strict unseen-domain/unseen-split (UDU) accuracy, the minimum receiver-level accuracy (receiver floor), and the strict floor under satellite-stress augmentation. The stress metrics remain proxy metrics and do not certify target adaptation.

C. Phase 2 Matched Matrix

The RTB-IDR diagnostic fixes five target receivers, seeds 713102–713106, and five slices:

$$(K, C_{\mathrm{n}}) \in \{(1,20), (5,20), (10,5), (10,10), (10,20)\}.$$

Each receiver–seed–slice row evaluates all three simulated LEO profiles, yielding 125 jobs and 375 scenario cells. The authoritative retry completed 125/125 jobs with zero execution failures.

The matched control, internally named D81, shares the frozen features, robust centers, dual-geometry fusion, safety gates, support/query sets, and seeds. The only controlled difference is whether Eq. (46) is activated after new-class registration. The paired deltas therefore isolate task-balanced covariance; they do not estimate the contribution of the entire RTB-IDR pipeline relative to a raw backbone.

TABLE V
Phase 1 Source-Only Results (%)

Metric	ADV2 avg.	CVS-RFFI P1	Difference
Overall	86.92	89.18	+2.26
Strict UDU	80.09	84.89	+4.80
Receiver floor	69.07	75.55	+6.48
Satellite strict floor	67.83	68.77	+0.94

D. External Comparisons

We include official-semantics reproductions of CSIL [15] and MoPC-HR [19]. Their complete matrix contains 5 receivers, 5 seeds, 4 values of K, 4 new-class scales, and two methods, for 800 cells and 2400 scenario rows. The new-class support and query records use the same simulated LEO channel convention, but the methods retain the base/source access required by their original training lifecycles. Their results are therefore descriptive, not paired with RTB-IDR.
[AUTHOR ACTION: Before submission, add same-capsule, same-seed, same-C_{n} support-only ProtoNet, single-qKNN, adapter-qKNN, and a frozen current CVS candidate. Report external-paper methods in a separate permission table rather than merging incompatible rows into one leaderboard.]

E. Evidence and Statistical Policy

All adaptations are constructed from support only. Hyperparameters are locked before query scoring. Each query prediction is written to an immutable artifact, and a separate scorer joins truth afterward. Main interpretations use complete same-row tuples; marginal maxima from different receivers or seeds are not combined. Means are descriptive unless a paired interval is available.
[AUTHOR ACTION: Compute receiver-clustered or receiver-by-seed bootstrap confidence intervals and paired effect intervals from the final frozen confirmation matrix. The current Phase 1 comparison has no independent-repeat interval and must not be described as statistically significant.]

VI. Results

A. Phase 1 Source-Only Generalization

Table V shows that the selected Phase 1 checkpoint improves the mean, strict split, and worst-receiver criteria simultaneously. The receiver-floor gain is the largest, suggesting that model selection based only on overall accuracy would understate the benefit. The satellite-stress floor improves by less than one point, however, and remains below a separate soft-unknown baseline's 69.55%. The selected checkpoint also has a source-overflow diagnostic of 0.459 and a bridge-accept diagnostic of 1.0.

These diagnostics prevent an open-set claim: the Phase 1 evidence supports closed-set source-only DG, not unknown rejection.

B. Matched Effect of Task-Balanced Covariance

Table VI yields three conclusions. First, the task-balanced covariance is inactive at $K = 1$, as required by the identifiability fallback; its paired effect is exactly zero. Second, as the new-class set grows, it increasingly protects old classes. At $K = 10, C_\mathrm{n} = 20$, all five target receivers improve in old-class accuracy, and the gains under clear, low-elevation, and rain stress are 2.00, 2.77, and 3.10 points. Third, the gain is not Pareto-complete. The same large-scale row loses 0.65 points of new-class accuracy, and the $K = 10, C_\mathrm{n} = 5$ row slightly favors new classes at the expense of old-class accuracy and floor.

The absolute values also matter. At $K = 10, C_\mathrm{n} = 20$, $H_{\mathrm{old,new}} = 69.56\%$, and the minimum old-class accuracy is 42.67%. Thus, complete execution and a positive paired delta do not imply deployment success.

C. Descriptive External Comparison

RTB-IDR obtains a substantially higher harmonic mean than the two reproduced incremental methods in Table VII. This is not a clean algorithm ranking. The old-pre values reveal that the methods enter registration with very different base representations, and their source/base permissions, training schedules, and seeds differ. The result instead shows that original incremental-learning recipes can become severely imbalanced when transferred to simulated LEO cross-receiver data with many simultaneously enrolled classes.

We intentionally exclude an older two-new-class qKNN table from this ranking. Changing C_n changes the candidate set and the cross-role confusion problem; comparing its H with a 5-, 10-, or 20-new-class row would be invalid.

D. Resource Profile

The compiled state is small, but registration is not free. The current $K = 10$ audit performs approximately 88 component fits and has an 11.15–11.74 GMAC-equivalent upper-bound accounting, together with FP64 covariance workspaces. The Python/NumPy/scikit-learn reference implementation decodes INT8 weights to FP32 before the final product. Accordingly, Table VIII supports a storage claim for the head, not an end-to-end INT8 acceleration claim.

VII. Discussion

A. What Is Novel and What Is Not

The individual ingredients of CVS-RFFI have precedents: Sinc filters [35], cosine-margin classification [36], domain-adversarial learning [24], style perturbation [25], shrinkage covariance [37], and Fisher discrimination [38]. The defensible novelty lies in their task-specific lifecycle and coupling:

1) an explicit separation between source-only semi-supervised receiver DG and support-only incremental deployment;
2) an immutable, quantized aggregate source-knowledge boundary rather than runtime source replay;
3) a class-permutation-symmetric state compiler that uses identical formulas for old and new supports, yet prevents new-class count from dominating covariance estimation; and
4) an evidence path in which every query is independently scored over all registered classes and truth is attached only after prediction publication.

The present evidence establishes these properties and a targeted forgetting reduction. It does not yet establish a state-of-the-art end-to-end accuracy result. For a top-tier Transactions submission, the protocol contribution must be accompanied by a stronger final Stage 2 method or by a broader benchmark release that makes the protocol itself independently reusable.

B. Failure Analysis

The $K = 1$ result exposes a structural limit: one support per class cannot identify within-class perturbation or estimate task-specific covariance. Query-assisted pseudo-labeling could add apparent samples, but it would violate the present support-only contract and would confound adaptation with evaluation.

At larger K, Eq. (46) balances covariance evidence, not score scales. New classes can still intrude into old-class decision regions. A licensed role-oracle diagnostic at $K = 10, C_\mathrm{n} = 20$ increases old accuracy from 71.33% to 83.31% and H from approximately 69.7% to 76.9%. The oracle is permanently non-promotable, but the gap localizes the remaining problem: legal cross-role calibration must be learned from support, not from a query's true role.

C. Validity Boundaries

Proxy-to-space gap: WiSig/ManySig is terrestrial. The simulated LEO operators are residual baseband proxies, not orbital measurements or a complete link-budget model. Claims about radiation, uncompensated Doppler dynamics, satellite payload electronics, rain attenuation, or flight reliability require real hardware and real links.

Selection multiplicity: Phase 1 was selected from 32 candidates. The reported comparison is an internal audit and lacks a fresh, independently repeated confirmation. Its deltas should not be interpreted as unbiased generalization gains until reproduced under a frozen selection rule.

External-method permissions: CSIL and MoPC-HR are reproduced according to their own lifecycles. Their low CVS results do not invalidate their original papers. A fair manuscript must show both original-task fidelity and the permission mismatch when transferring them to CVS.

Resource evidence: The final head is compact, but the complete registration pipeline has no worst-case execution

TABLE VI
RTB-IDR Absolute Results and Paired Change Versus the Matched Control

K	C_n	A_o^{pre}	A_o^{post}	Min-old	A_n	$H_{old,new}$	F	ΔA_o^{post}	ΔMin-old	ΔA_n	$\Delta H_{old,new}$
1	20	68.144	44.033	14.200	27.150	33.410	24.111	0.000	0.000	0.000	0.000
5	20	81.267	63.711	33.200	58.883	60.955	17.556	+2.311	+2.400	-0.410	+0.920
10	5	86.111	76.189	49.800	74.133	74.803	9.922	-0.133	-0.867	+0.520	+0.197
10	10	86.111	72.533	44.200	66.353	69.106	13.578	+1.000	+1.933	-0.340	+0.291
10	20	86.111	71.333	42.667	68.150	69.555	14.778	+2.622	+4.600	-0.653	+0.964

TABLE VII
Different-Permission Comparison at $K = 10, C_n = 20$ (%)

Method	Old-pre	Old-post	New	H
RTB-IDR	86.111	71.333	68.150	69.555
CSIL reproduction	42.833	38.222	1.660	2.979
MoPC-HR reproduction	45.322	7.611	25.187	10.695

TABLE VIII
RTB-IDR Compiled-State Resource Profile

C	FP32 affine	Core state	Head MAC/query
6	6.77 KiB	4.58 KiB	1,728
11	12.42 KiB	7.46 KiB	3,168
16	18.06 KiB	10.34 KiB	4,608
26	29.35 KiB	16.11 KiB	7,488

TABLE IX
Evidence Lock for the Submission Version

Claim surface	Current status and required evidence
Phase 1 DG	Internal audit only; add fresh frozen repeats, intervals, module ablations, and matched cross-receiver baselines.
Stage 2 registration	Negative diagnostic; add a frozen promotable method on the full K, receiver, seed, scenario, and C_n matrix with same-row metrics.
Support baselines	Existing slices differ; match capsule, seeds, new-class scale, physical IDs, and permission rows for qKNN/ProtoNet.
Satellite transfer	Untested; add real-satellite or hardware-in-the-loop validation, or retain proxy-only language.
Onboard efficiency	Head storage only; measure end-to-end latency, peak RAM, energy, and numerical closure on the target processor.
Reproducibility	Local evidence only; release code, hashes, split manifest, channel parameters, and a truth-isolated scorer.

time, peak resident memory, energy, thermal, or radiation-tolerance measurement on a target processor.

Table IX separates the current evidence lock from the measurements required for a Q1 submission. In particular, neither compact head storage nor completed diagnostic execution substitutes for end-to-end onboard validation or a frozen promotable Stage 2 result.

VIII. Conclusion

This paper introduced CVS-RFFI, a protocol-governed framework for cross-receiver RFFI under a space-borne deployment proxy. It separates source-only dual-representation learning from support-only old-class adaptation and new-class enrollment, and it makes the deployment boundary auditable through fixed received observations, immutable model knowledge, independent all-class decisions, and truth-side scoring. The current RTB-IDR realization compiles multimodal target support into a compact discriminant state. Its 125-row matched diagnostic shows that task-balanced covariance can reduce old-class forgetting when many new classes are registered, but the improvement is accompanied by a new-class tradeoff and vanishes at one shot. The evidence therefore supports the protocol, the state-compilation mechanism, and a specific forgetting result; it does not yet support in-orbit validation or a completed state-of-the-art deployment claim. Resolving support-only cross-role calibration, one-shot identifiability, and target-hardware efficiency is the necessary next step.

Data and Code Availability

The base WiSig dataset is described in [7]. The current experiment artifacts, protocol capsules, and implementation are maintained in a controlled research repository. [AUTHOR ACTION: Replace this paragraph with the final public repository, immutable release tag, data-access instructions, split-manifest hashes, and artifact archive DOI. Do not promise release terms that have not been approved.]

Acknowledgment

[AUTHOR ACTION: Insert funding, compute-resource, data-provider, and contributor acknowledgments after authorship review.]

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